

Module 1 — Quiz B (Concept Check)

Question bank · After Read 2 · open notes · unlimited attempts

Module 1, Quiz B: Concept check

Reading: [Module 1 Read 2](#) (complete Read 1 + Quiz A first)

Canvas: Concept Check quiz (Module 1)

Binomial **test** skills from Read 2: translate research questions to H_0 and H_a ; identify n , null p , and observed x in context; check BRP conditions; read null distributions and cumulative tails; interpret p-values and conclusions; one- vs two-sided shading. **R** and **ggplot workflow** items are on the [Synthesis](#) bank after the [demo lab](#).

Excluded from Module 1: Bayes' rule — optional enrichment only (not on the course path).

Section A — Research questions and hypotheses (Read 2 §1)

A1. Left-handed students

At a school, **25 students** are surveyed. The national rate of left-handedness is **10%**. The research question is: *Is the proportion of left-handed students at this school **higher** than 10%?*

Let p = proportion of students at this school who are left-handed.

1. Write H_0 and H_a in symbols.
2. In one sentence each: what does H_0 claim? What does H_a claim?

A2. ESP test hypotheses

An online ESP test uses **5 symbols** per trial. A student takes **25 trials**. Let p = probability of a correct guess on **one** trial.

Write H_0 and H_a for the claim “*The student has ESP (does better than guessing).*”

A3. Describe p in context

A quality team tests whether the proportion of **defective items** from a new supplier is **less than 15%**. They inspect **25 items**.

In **one complete sentence**, define the parameter p in the context of this study.

A4. Null vs alternative

Which statement belongs in the **null hypothesis** H_0 for a binomial test about a proportion p ?

- a) $p > 0.20$
- b) $p = 0.20$
- c) $p \neq 0.20$
- d) The data show p is too high

A5. Two-sided coin test

A student claims a coin is **not fair** (not 50% heads). They flip it **20 times** and count heads. Let p = probability of heads on one flip.

Write H_0 and H_a for a **two-sided** test.

A6. Gym use (harder)

Research question: *Do **more than 20%** of students at our school use the gym at least once per week?* A random sample of **30 students** is surveyed; **9** say yes.

1. Write H_0 and H_a .
2. Is this one-sided or two-sided? Which direction?

Answer key — Section A — Research questions and hypotheses (Read 2 §1)

1. $H_0 : p = 0.10$ vs $H_a : p > 0.10$. H_0 : the school rate equals the national 10%. H_a : the school rate is greater than 10%.
 2. $H_0 : p = 0.2$ (or $1/5$) vs $H_a : p > 0.2$. Under H_0 the student is guessing; under H_a they beat chance.
 3. p is the proportion of items from this supplier that are defective (on one item / in the long run for this process).
 4. **b)** H_0 always uses an **equality** (here $p = 0.20$). Inequalities go in H_a .
 5. $H_0 : p = 0.5$ vs $H_a : p \neq 0.5$. Two-sided because “not fair” means different from 0.5 in either direction.
 6. $H_0 : p = 0.20$ vs $H_a : p > 0.20$. **One-sided**, right tail (looking for evidence the rate exceeds 20%).
-

Section B — Identify n , x , and p (Read 2 §1–2)

B1. n and x from a word problem

A researcher surveys **30 students**; **12** report owning a bicycle. The binomial count is “owns a bicycle” on each student.

1. What is n ?
2. What is the observed count x ?
3. What is $\hat{p}$?

B2. p under H_0

Scenario: At a school, **25 students** are surveyed. The national rate of left-handedness is **10%**. You test $H_0 : p = 0.10$ vs $H_a : p > 0.10$ and will build the **null distribution** $f(x) = P(X = x)$ under H_0 with $n = 25$ trials.

What value of p belongs in the binomial formula for that null bar chart? Why (one short phrase)?

B3. Observed x in ESP test

ESP test: **25** trials, **6** correct guesses. Under $H_0 : p = 0.2$:

1. What is n ?
2. What is the observed x ?
3. What is $\hat{p}$?

B4. $\hat{p}$ vs null p (harder)

Scenario: A quality team tests whether the proportion of **defective items** from a supplier is **less than 15%**. They inspect $n = 25$ items. $H_0 : p = 0.15$. They observe **3** defectives, so $\hat{p} = 3/25 = 0.12$.

1. Which value (**0.12** or **0.15**) goes into the formula for **null bar heights** $P(X = x)$ under H_0 ?
2. Which value describes **what you observed** in the sample?

B5. X vs x

Scenario: You plan $n = 25$ independent trials and let X = number of successes before data are collected. After data collection you observe **8** successes.

1. Which symbol is the **random variable**?
2. Which value is the **observed count** you mark on the null plot with a vertical line?

B6. Fixed n before data

Scenario: Before collecting data, you decide to run exactly **25 independent trials** (for example, 25 ESP guesses or 25 item inspections). You do not stop early or add trials based on results.

Why does fixed n matter for a binomial test? Which BRP condition does it support?

B7. Identify all three (harder)

Basketball player: *Is her free-throw percentage **greater than 70%**?* She takes **20** shots and makes **16**.

Let p = probability she makes **one** free throw (long-run). State n , observed x , null p from H_0 , and $\hat{p}$.

Answer key — Section B — Identify n , x , and p (Read 2 §1–2)

1. $n = 30$; $x = 12$; $\hat{p} = 12/30 = 0.4$.
 2. $p = 0.10$ — use the **null** value from H_0 , not the observed $\hat{p}$ from data.
 3. $n = 25$; $x = 6$; $\hat{p} = 6/25 = 0.24$.
 4. Null bar chart: $p = 0.15$ from H_0 . Observed: $\hat{p} = 0.12$ (or $x = 3$).
 5. X is the random variable (before data); $x = 8$ is the observed count on the plot.
 6. You must know the number of trials before collecting data; supports the **fixed number of trials** condition in a BRP.
 7. $n = 20$; $x = 16$; null $p = 0.70$ under H_0 ; $\hat{p} = 16/20 = 0.8$.
-

Section C — BRP conditions in context (Read 2 §1–2)

C1. State the four conditions

List the **four BRP conditions** in words (binary, fixed n , fixed p , independent trials).

C2. Drawing cards without replacement

You draw **10 cards without replacement** from a deck and count kings. Which BRP condition **fails**? Explain in one sentence.

C3. ESP open deck

ESP test: 25 trials, 5 symbols, computer randomizes each trial independently; open deck so each symbol has probability $1/5$ every trial.

Name **two** BRP conditions and explain briefly why each holds here.

C4. Survey without replacement

A class has **40 students**. You sample **10 without replacement** and count how many commute by bike. Is this a standard BRP for testing a population proportion? **Yes or no**, and one reason.

C5. Independent coin tosses

Fair coin tossed **15 times**; count heads. For a binomial model, explain why **independent trials** is reasonable.

C6. Which study is binomial? (harder)

Which setup is **best** modeled as a BRP for a binomial test?

- a) Flip a bent coin 20 times; the bend gets worse after each flip.
- b) 25 independent ESP guesses with fixed chance 0.2 under H_0 .
- c) Count rain days in July until the first dry day, then stop.

Answer key — Section C — BRP conditions in context (Read 2 §1–2)

1. Binary outcome each trial; fixed number of trials n ; same success probability p on each trial; independent trials.
 2. **Fixed** p (and often independence): the probability of a king changes as the deck changes.
 3. e.g. **Binary** (correct/incorrect); **fixed** $n = 25$; **fixed** $p = 0.2$ under H_0 ; **independent** trials (random each time).
 4. **No** (or only approximate): sampling without replacement changes p slightly between draws; trials are not fully independent with fixed p .
 5. One toss does not change the chance of heads on the next; outcomes are separate random trials.
 6. b) Fixed n , binary, fixed p , independent trials. (a) p changes; (c) n not fixed.)
-

Section D — Null distribution, graphs, and cumulative probability (Read 2 §2–3)

D1. Most likely value of x

Under H_0 , $X \sim \text{Binomial}(n = 10, p = 0.5)$. The null bar chart shows $P(X = x)$ for each x .

At which x is the **tallest bar** (most likely count)? **Why** (one phrase: mean or symmetry)?

D2. Center of null plot

Under $H_0 : p = 0.2$, $n = 25$, where is the **center** of the null distribution approximately? Compute $\mu = np$.

D3. $P(X \geq x)$ as sum of bars

Null distribution for $n = 10$, $p = 0.4$. You shade the **right tail** for $H_a : p > 0.4$ at observed $x = 7$.

Write $P(X \geq 7)$ as a **sum** of pmf values $f(7) + f(8) + \dots$ (use $f(x)$ notation).

D4. Lower tail cumulative

Scenario: Defective items: $H_0 : p = 0.15$, $H_a : p < 0.15$, $n = 25$ items inspected, observed **3 defectives**. The p-value uses the **left tail**.

Write the p-value as $P(X \leq 3)$ using a sum of bars. Which x values are included?

D5. Two-sided sum (harder)

Two-sided test: $H_0 : p = 0.5$, $n = 20$, observed $x = 3$ (unusually low). Name the **two sets of x values** whose bar heights you would add for a two-sided p-value (qualitative — no calculation).

D6. Reading a null plot

A null bar chart under H_0 shows $f(4)$ is the **largest** bar and bars taper for $x = 0, 1, 2, 3$ and $x = 5, 6, \dots$

If you observe $x = 8$, is that toward the **right tail** relative to the peak? **Yes or no.**

D7. Bar height on null plot

Scenario: Under $H_0 : p = 0.20$ with $n = 25$ trials, the null bar chart shows $f(x) = P(X = x)$ at each x .

What does $f(4)$ represent on that plot?

Answer key — Section D — Null distribution, graphs, and cumulative probability (Read 2 §2–3)

1. $x = 5$ (center at $np = 5$; symmetric when $p = 0.5$).
 2. $\mu = np = 25(0.2) = 5$ — counts near 5 are most likely under H_0 .
 3. $P(X \geq 7) = f(7) + f(8) + f(9) + f(10)$ where $f(x) = P(X = x)$ under H_0 .
 4. $P(X \leq 3) = f(0) + f(1) + f(2) + f(3)$ under H_0 with $p = 0.15$.
 5. Counts as extreme as 3 toward **low** (0, 1, 2, 3 or ≤ 3) **and** counts equally extreme on the **high** side (mirror: ≥ 17 if symmetric rule, or both tails away from center).
 6. **Yes** — 8 is far to the right of the peak at 4 (right-tail direction for “more successes than null expects”).
 7. $f(4) = P(X = 4)$ — the **height** of the bar at $x = 4$ under the null model.
-

Section E — p-value meaning and conclusions (Read 2 §2–3)

E1. Small p-value

Scenario: A binomial test about whether a school’s left-handed rate exceeds 10% gives **p-value = 0.02** for $H_a : p > 0.10$.

In one sentence, what does that mean **assuming** H_0 **is true**?

E2. Large p-value

Scenario: ESP test with **5 symbols**, **25 trials**, under $H_0 : p = 0.2$. P-value = **0.76** for $H_a : p > 0.2$.

Do the data provide **strong evidence** for ESP? **Fail to reject** or **reject** H_0 ? One sentence conclusion.

E3. Common mistake

A student says: “The p-value is 0.04, so there is a **4% chance** H_0 **is true**.” Is that correct? If not, give the correct interpretation in one sentence.

E4. Interpret in context

Left-handed study: $H_0 : p = 0.10$, $H_a : p > 0.10$, $n = 25$, observed $x = 6$, p-value = **0.03**. Write a conclusion about the research question (not “accept H_a ” as proof).

E5. Compare p-values (harder)

Scenario: Two studies use the same H_0 , H_a , and n . Study A: p-value = **0.001**. Study B: p-value = **0.45**.

Which provides **stronger evidence against** H_0 ? Which observed count is **more unusual** under H_0 ?

E6. Conclusion language

Which conclusion is **appropriate** after p-value = 0.08?

- a) Accept H_0 as proven true.
- b) Fail to reject H_0 ; data are consistent with H_0 .
- c) Accept H_a with 92% certainty.

E7. Right-tail p-value setup

Gym study: $H_0 : p = 0.20$, $H_a : p > 0.20$, $n = 30$, $x = 9$. Write the p-value as a probability statement about X under H_0 .

E8. Full mini-conclusion (harder)

Defective items: $H_0 : p = 0.15$, $H_a : p < 0.15$, $n = 25$, $x = 3$, p-value = **0.22**. Is the defect rate convincingly **below 15%**? One-sentence conclusion.

Answer key — Section E — p-value meaning and conclusions (Read 2 §2–3)

1. If H_0 were true, there is only a **2%** chance of seeing a count as or more extreme as the observed data toward H_a .
 2. No strong evidence for ESP; **fail to reject** H_0 — the observed count is plausible if guessing only.
 3. **Incorrect.** The p-value is the chance of data **at least this extreme given** H_0 , not $P(H_0 \text{ true})$.
 4. There is statistically significant evidence the school's left-handed rate **exceeds 10%** (unusual under H_0); reject H_0 at this level.
 5. Study A (0.001) — smaller p-value, more unusual under H_0 .
 6. **b)** We never “accept H_0 ” or “prove” either hypothesis — insufficient evidence to reject H_0 .
 7. $p\text{-value} = P(X \geq 9 \mid H_0)$ with $X \sim \text{Binomial}(30, 0.20)$.
 8. **No** — fail to reject H_0 ; p-value 0.22 is large, so 3 defectives (or fewer) is not strong evidence the rate is below 15%.
-

Section F — Test workflow and tail direction (Read 2 §3)

F1. Order the steps

Number these binomial-test steps **1–4** in the order used in Read 2:

- **(A)** Verify the four BRP conditions.
- **(B)** State research question, p , H_0 , H_a ; identify n and null p .
- **(C)** Plot null distribution (bar chart of $P(X = x)$ under H_0).
- **(D)** Collect x , mark observed value on the plot, shade tail(s), compute p-value, conclude.

F2. One- vs two-sided shading

1. $H_a : p > 0.2$: shade **one** or **both** tails?
2. $H_a : p \neq 0.5$: shade **one** or **both** tails?

F3. Which tail?

Basketball: $H_0 : p = 0.70$, $H_a : p > 0.70$, observed $x = 18$ of $n = 20$. For the p-value, do you sum bars for $P(X \geq 18)$ or $P(X \leq 18)$?

F4. P-value as probability

Scenario: ESP test: $H_0 : p = 0.20$, $H_a : p > 0.20$, $n = 25$ trials, observed $x = 7$ correct guesses.

Which expression equals the **p-value** (assuming H_0 is true)?

F5. Theory before data (harder)

Scenario: Before your Project 1 data arrive, you draw the milestone **null bar chart** under H_0 (you do not yet know the final observed x).

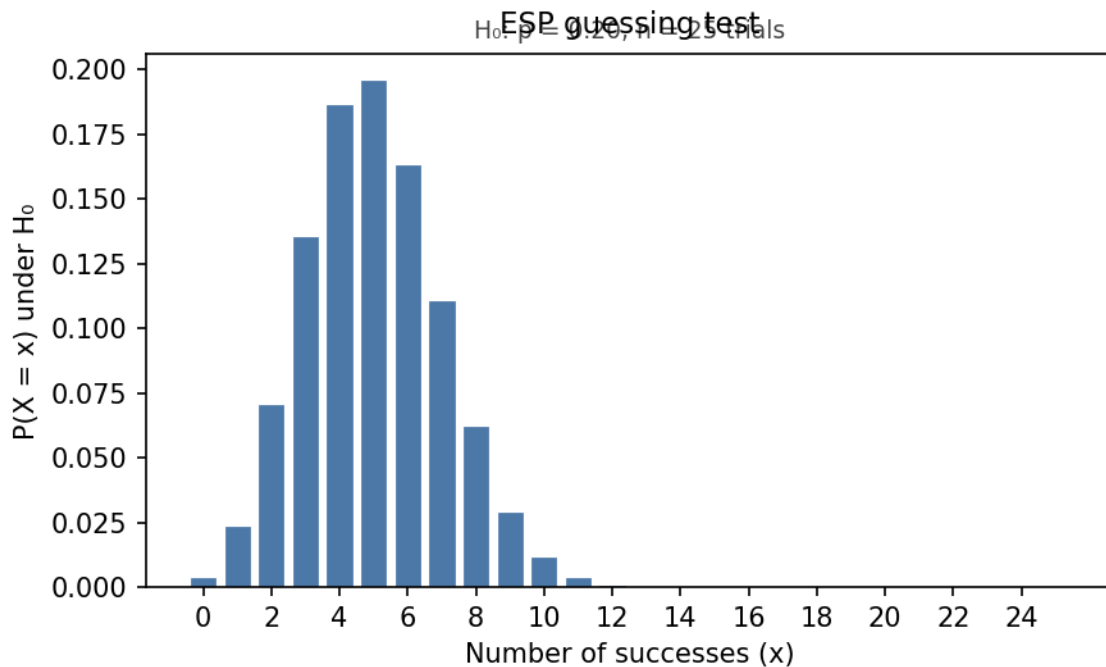
Why should that null plot be drawn **before** you know your final observed x ? Tie to the “world we imagine” idea in one or two sentences.

Answer key — Section F — Test workflow and tail direction (Read 2 §3)

- 1=B, 2=A, 3=C, 4=D (plan before data; null plot before observed x).
 1. **One** tail (right). 2. **Both** tails.
 - $P(X \geq 18)$ — **right tail** matching $p > 0.70$.
 - $P(X \geq 7)$ where $X \sim \text{Binomial}(25, 0.20)$ — right tail toward H_a .
 - Good science plans the model under H_0 first; the null plot describes what counts are typical **before** seeing data, avoiding picking H_0 after x .
-

Section G — Null distribution plots (Read 2 §2–3)

G1. ESP null — most likely count

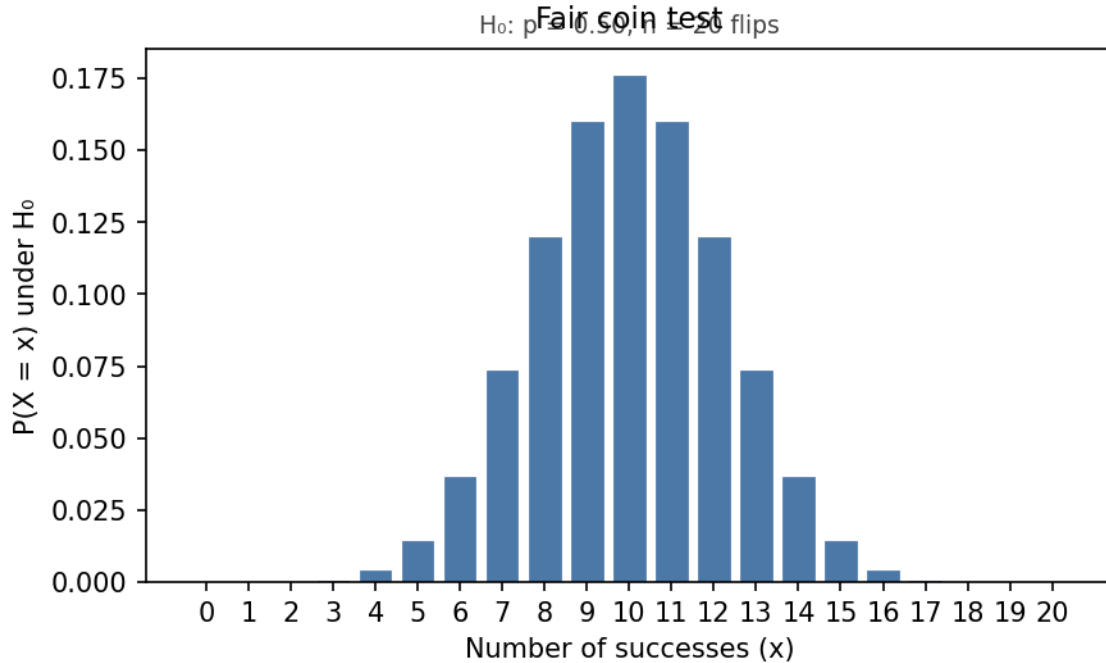


Scenario: An online ESP test uses **5 symbols** per trial. Under the null hypothesis $H_0 : p = 0.20$ (random guessing), you plan $n = 25$ independent trials and let $X =$ number of correct guesses.

The figure shows the **null distribution** — bar height at each x is $P(X = x)$ under H_0 .

At which value of x is the **tallest bar** (most likely count if H_0 is true)?

G2. Coin null — most likely count

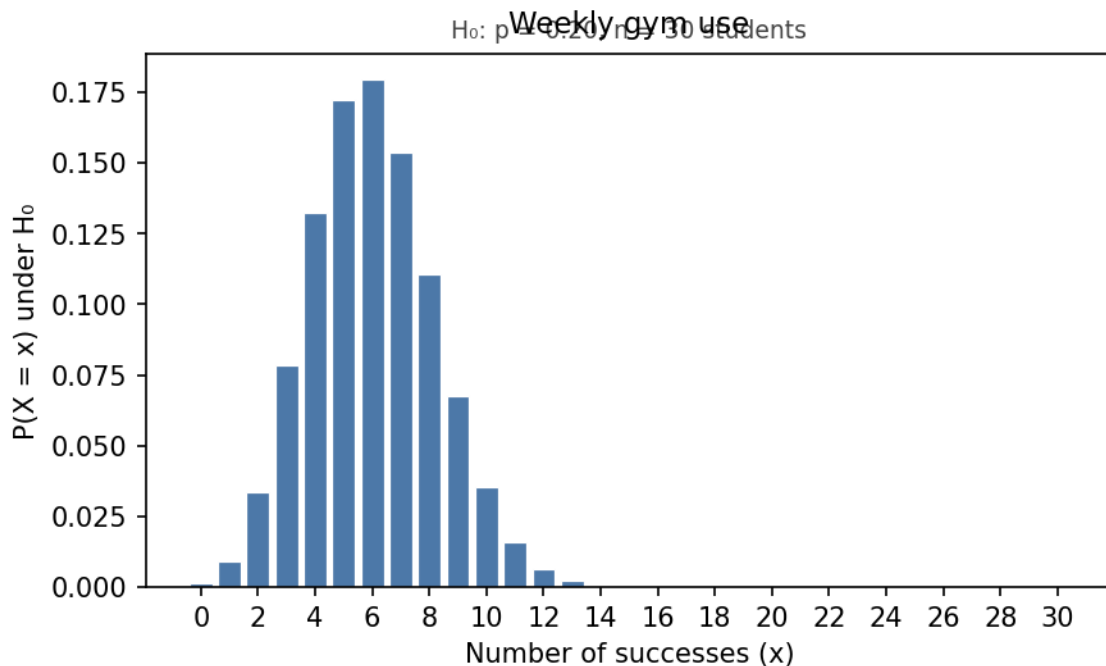


Scenario: A student tests whether a coin is fair. Under $H_0 : p = 0.50$ (probability of heads on one flip), they will flip the coin $n = 20$ times and count heads.

The figure shows the null distribution under H_0 .

Which x has the **highest** bar?

G3. Gym null — center

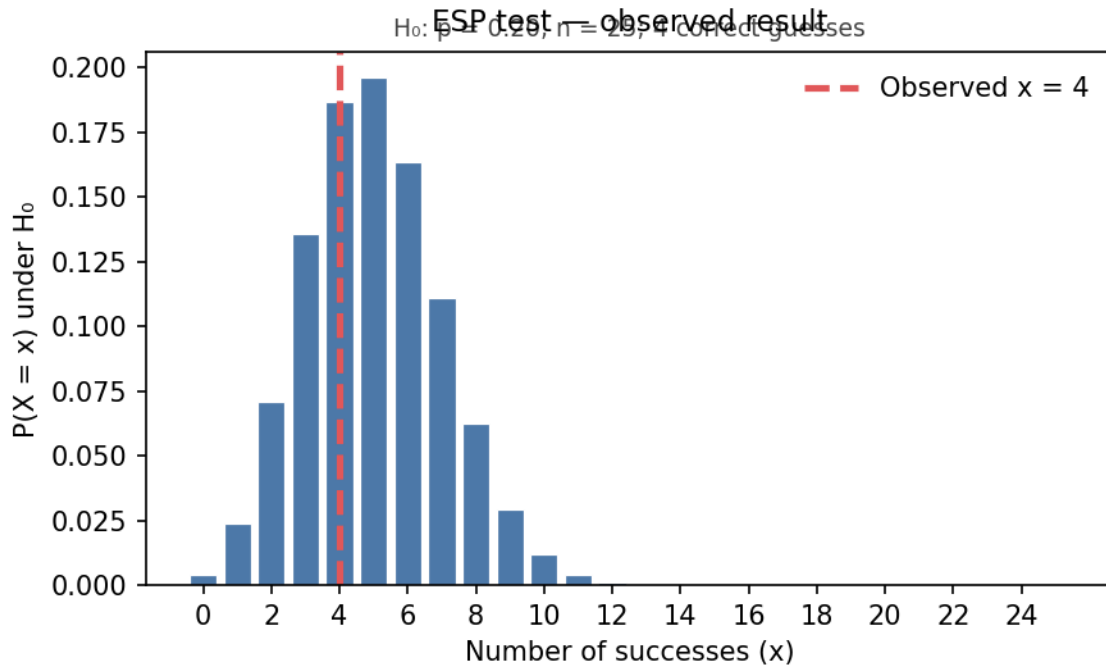


Scenario: A school surveys $n = 30$ students about weekly gym use. Under $H_0 : p = 0.20$ (20% use the gym), let X = number who say yes.

The figure shows the null distribution.

Approximate the **center** of the distribution ($\mu = np$). Which x is nearest the tallest bar?

G4. ESP — $x = 4$ vs null

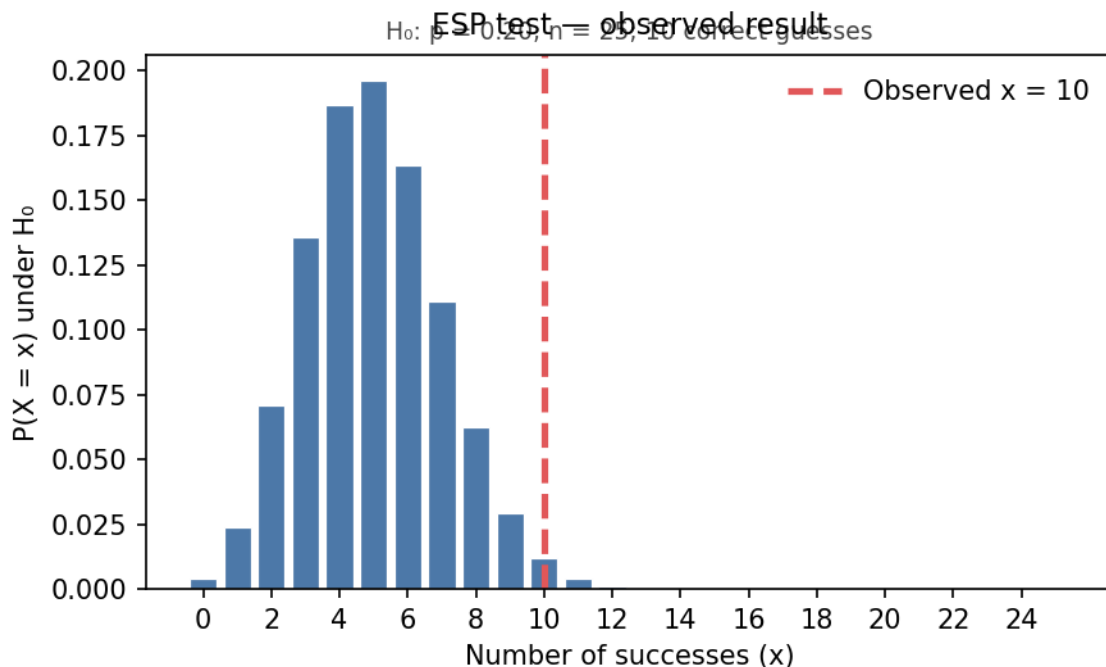


Scenario: ESP test with $n = 25$ trials and **5 symbols** (guessing gives $H_0 : p = 0.20$). A student gets **4 correct** guesses ($x = 4$).

The figure shows the null distribution with a **dashed vertical line** at the observed $x = 4$.

Is $x = 4$ **consistent with** H_0 (plausible near the center of the null plot)? **Yes or no**, with one phrase.

G5. ESP — $x = 10$ vs null

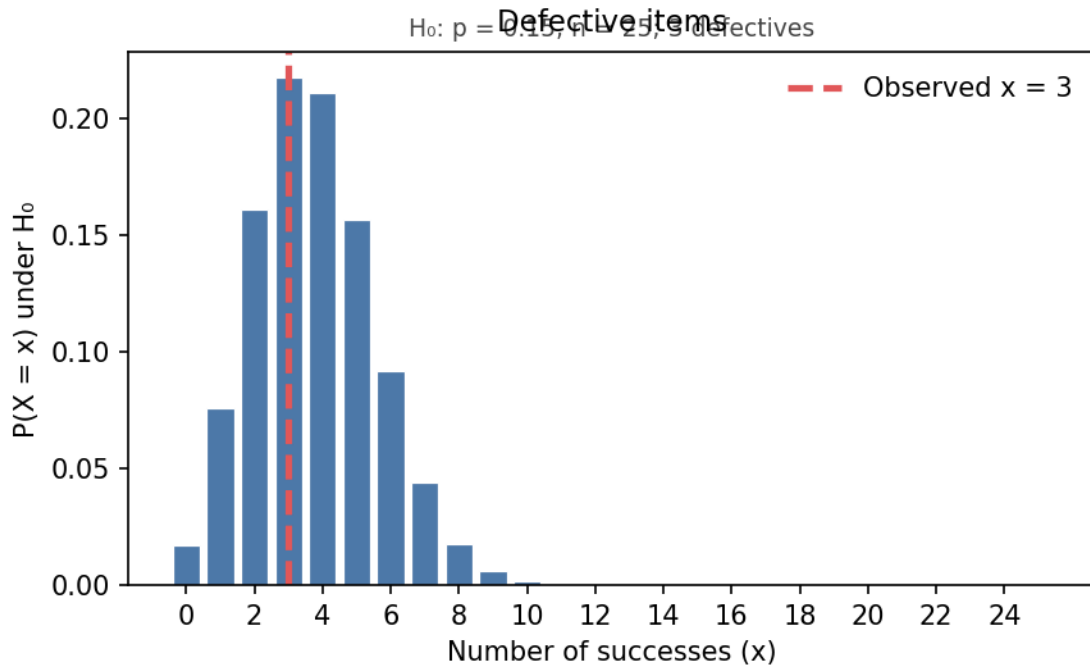


Scenario: An online ESP test uses **5 symbols** per trial. Under $H_0 : p = 0.20$ (random guessing), $H_a : p > 0.20$, you run $n = 25$ independent trials. Observed $x = 10$ correct guesses.

The figure shows the null distribution with a line at $x = 10$.

Relative to the null distribution, is $x = 10$ in the **right tail** (unusually high if H_0 were true)? **Yes or no.**

G6. Defectives — $x = 3$ vs null

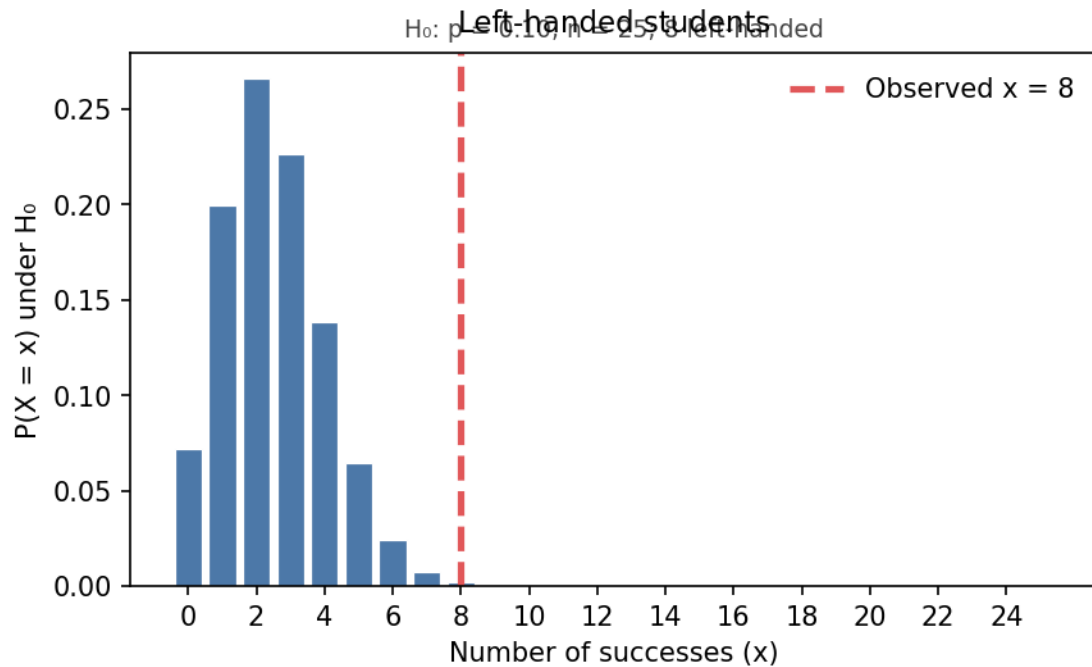


Scenario: A quality team tests whether the defect rate is **below 15%**. $H_0 : p = 0.15$, $H_a : p < 0.15$, $n = 25$ items inspected, $x = 3$ defectives.

The figure shows the null distribution with a line at $x = 3$.

Is the observed count **plausible** under H_0 (not far in the left tail)? **Yes or no.**

G7. Left-handed — $x = 8$ vs null



Scenario: 25 students surveyed; national left-handed rate is 10%. $H_0 : p = 0.10$, $H_a : p > 0.10$. Observed $x = 8$ left-handed students.

The figure shows the null distribution with a line at $x = 8$.

Is $x = 8$ toward the **right tail** of the null plot (suggesting evidence for H_a)? **Yes or no.**

Answer key — Section G — Null distribution plots (Read 2 §2–3)

1. The tallest bar is at $x = 5$ (center at $\mu = np = 25 \times 0.2 = 5$).
 2. $x = 10$ (symmetric null with $np = 10$).
 3. Center $\mu = 30 \times 0.2 = 6$; the tallest bar is near $x = 6$.
 4. **Yes** — 4 is near the center ($np = 5$); not extreme for H_0 .
 5. **Yes** — 10 is far to the right of the peak near 5; counts this high are rare under H_0 .
 6. **Yes** — 3 is on the left side but still a reasonably likely count under $p = 0.15$.
 7. **Yes** — 8 is far above the center ($np = 2.5$); unusually high if H_0 were true.
-